ODD+D Appendix for Searching Legislative Design Space

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This appendix documents the congressional institutional simulator using the ODD and ODD+D model-description conventions [Grimm et al. 2010; Müller et al. 2013]. It describes entities, state variables, scheduling, adaptation, input data, submodels, outputs, calibration screening, assumptions, and reproducibility commands for the simulator used in *Searching Legislative Design Space for Compromise and Productivity.*

CCS Concepts: • **Computing methodologies** → **Modeling and simulation**; • **Human-centered computing** → *Collaborative and social computing theory, concepts and paradigms.*

Additional Key Words and Phrases: ODD+D, legislative simulation, institutional design, model documentation

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1 Purpose

The model compares legislative institutional designs under shared generated political worlds. Its purpose is to search for mechanisms that improve compromise and productivity while preserving representative responsiveness, public-benefit alignment, minority protection, and resistance to capture. It is not a forecast of a specific Congress. It is a mechanism-search simulator: the same generated legislators and bills are routed through many institutional processes so differences can be attributed to procedural rules and incentive structures.

2 Entities, State Variables, and Scales

2.1 Legislators

Legislators have an identifier, party label, ideology, compromise preference, party loyalty, constituency sensitivity, lobbying susceptibility, reputation sensitivity, district preference, district intensity, and affected-group sensitivity. These values determine how the legislator weighs ideology, party pressure, constituent/public support, lobbying pressure, compromise incentives, reputation, and concentrated harm before casting a final yes-or-no vote.

2.2 Bills

Bills have an identifier, title, proposer, proposer ideology, original and current policy position, public support, generated public benefit, public-benefit uncertainty, lobby pressure, private gain, salience, issue domain, anti-lobbying-reform flag, affected group, affected-group support, concentrated harm, compensation cost, cosponsor counts, amendment movement, lobbying channel spend, challenge status, citizen-review signals, active-law signals, and alternative-selection signals.

2.3 Lobby Groups

Lobby groups are explicit collective actors. Each group has an identifier, primary issue domain, issue-preference map, preferred policy position, budget, influence intensity, defensive multiplier, information bias, public-campaign skill,

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capture strategy, and public-support mismatch tolerance. Groups can spend through direct pressure, agenda-access pressure, information distortion, public campaigns, litigation or delay threats, and defensive anti-reform pressure.

2.4 Institutional Processes

An institutional process is a composable module. Processes can screen proposal access, apply leadership agenda control, modify public signals, spend lobbying or attention resources, route bills to procedural tracks, mediate amendments, evaluate affected-group harm, select among alternatives, conduct chamber votes, revise through conference committees, apply vetoes, apply judicial review, route through citizen initiatives, register enacted laws, or review laws later. The final decision still resolves to a yes-or-no outcome, enactment or rejection, and a post-bill status quo.

3 Process Overview and Scheduling

Each run follows a fixed schedule:

(1) Generate a world containing legislators, party positions, lobby groups, a status quo, and a bill stream.
(2) Build each scenario's legislative process from that world.
(3) For each bill, create a deterministic vote context containing party positions, status quo, and seeded randomness.
(4) Pass the bill through the scenario process chain.
(5) Record the bill outcome, including agenda disposition, vote totals, enactment, amendments, challenges, lobbying signals, public-review signals, and policy movement.
(6) Update the scenario's status quo when policy is enacted or rolled back.
(7) Aggregate outcomes into scenario metrics.

Campaigns repeat this over many runs and assumption cases. Scenarios share generated worlds but use independent deterministic random streams, supporting paired comparison among institutions.

4 Design Concepts

4.1 Basic Principles

The model assumes legislative outcomes are shaped by proposal access, agenda scarcity, voting thresholds, committee and information structures, lobbying and capture, amendment opportunities, public review, concentrated harm, reversibility, and strategic actor adaptation.

4.2 Emergence

Productivity, compromise, capture, volatility, legitimacy, and gridlock emerge from bill-by-bill interactions. No actor directly optimizes the reported global metrics.

4.3 Adaptation

Proposers adapt over repeated bills. They can reduce proposal volume, delay timing, moderate risky bills, seek outside-bloc cosponsorship, reduce lobby exposure, add compensation amendments, lower risk tolerance after bad outcomes, and withdraw weak risky bills. Proposer state includes trust, proposal pace, risk tolerance, concession rate, lobby exposure, cooldown, and streak counters.

Lobby groups adapt over repeated bills. They track returns by channel, select capture strategies from learned returns and current bill conditions, change issue-specific spending multipliers, adjust overall budget intensity, increase reform-threat sensitivity when anti-lobbying bills become threatening, and maintain defensive pressure against reforms that would reduce lobbying influence.

Institutional state also adapts through proposal credits, proposal bonds, law registries, sunset review, challenge-token exhaustion, attention-token spending, norm erosion, judicial reversals, and active-law correction.

4.4 Objectives

Legislators vote according to weighted incentives. Proposers seek agenda access and enactment while learning to avoid reputational or procedural failure. Lobby groups seek policy movement toward their issue preferences and private gains while resisting anti-lobbying reforms. Institutions do not optimize a single welfare function; they impose constraints and incentives whose tradeoffs are measured afterward.

4.5 Learning and Prediction

Learning is deterministic and bounded rather than a full reinforcement-learning procedure. Actors use local outcome signals: enactment, public support, public benefit, capture risk, concentrated harm, salience, private gain, prior failures, and prior channel returns.

4.6 Stochasticity

World generation, committee composition, citizen panels, bill signals, party-system profiles, and some routing decisions use seeded randomness. Identical seeds reproduce identical outputs.

4.7 Observation

The simulator reports productivity, floor load, enacted support, public-benefit proxy, cooperation, compromise, gridlock, access denial, committee rejection, challenge rate, low-support passage, policy shift, proposer gain, lobby capture, public alignment, anti-lobbying success, private-gain ratio, lobbying spend by channel, public benefit per lobby dollar, amendment rate, amendment movement, minority harm, concentrated-harm passage, compensation rate, legitimacy, active-law welfare, reversal rate, correction delay, low-support active-law share, selected-alternative distance, proposer agenda advantage, alternative diversity, status-quo win rate, citizen-review metrics, attention spending, objection-window use, public-will review, district alignment, cosponsorship, bond forfeiture, strategic decoys, proposer-access Gini, welfare per submitted bill, vetoes, and overrides.

5 Initialization

`WorldGenerator` creates legislators from party-system and ideology parameters, derives party positions, generates lobby groups by issue domain, chooses a scalar status quo, and generates bills around proposer ideal points. Generated values are bounded by model validation. Party-system profiles include ideological bins, two major parties with minor parties, fragmented multiparty systems, dominant-party systems, and weighted campaign profiles.

6 Input Data and Calibration

Normal scenario runs use synthetic generated worlds. Calibration screening reads the tracked benchmark extract at `data/calibration/empirical-benchmarks.csv`. The extract maps empirical sources to simulator metrics and

benchmark ranges. Current named sources include Voteview roll-call data [Lewis et al. 2021], Congress.gov and govinfo bill histories [Library of Congress 2026; United States Government Publishing Office 2020], Comparative Agendas Project topic data [Comparative Agendas Project 2026], ParlGov party-system data [Döring and Manow 2024], U.S. Lobbying Disclosure Act filings [United States Senate Office of Public Records 2026], Center for Effective Lawmaking scores [Center for Effective Lawmaking 2026; Volden and Wiseman 2014], and V-Dem institutional indicators [V-Dem Institute 2026].

The calibration runner executes conventional scenarios and writes CSV and Markdown pass/fail reports under `reports/`. The current calibration pass is an empirical screen, not an automatic parameter fitter. Its benchmark ranges are intentionally broad and several are abstract proxies, so passing the screen should be read as minimum plausibility rather than validation. Its purpose is to keep ordinary baselines within explicit plausible ranges before interpreting counterfactual mechanisms.

7 Submodels

7.1 Voting

Legislators cast `YAY` or `NAY` through weighted influences: ideology, party, constituency/public support, lobbying, compromise preference, reputation, and affected-group sensitivity.

7.2 Proposal Access and Agenda Scarcity

Access rules include open access, viability screens, scalar proposal costs, public-value waivers, lobby surcharges, member quotas, leadership agenda control, public-interest screens, cross-bloc cosponsorship gates, affected-group sponsor gates, proposal bonds, earned proposal credits, agenda lotteries, and quadratic attention budgets.

7.3 Committees and Information

Committee gatekeeping can reject proposals before floor consideration. Committee information processes move public-signal estimates toward generated public benefit with composition-dependent accuracy.

7.4 Challenge and Objection

Challenge vouchers allocate scarce blocking rights to parties or members. Q-member escalation lets a sufficient number of legislators trigger active review. Public objection windows route contested bills to active voting, higher thresholds, or repeal review depending on scenario.

7.5 Lobbying

Budgeted lobbying modifies bill pressure and public signals through channel-specific spending. Anti-capture variants include transparency, public financing, public advocates, blind review, audits, sanctions, and defensive-spend caps. Adaptive lobbying learns from channel returns and reform threats.

7.6 Amendment and Mediation

Structured mediation moves risky proposals toward weighted institutional targets. Multi-round mediation models repeated amendments, costs, concession limits, poison-pill risk, and compensation for concentrated harm.

7.7 Distributional Harm

Harm-weighted thresholds, compensation amendments, and affected-group consent mechanisms distinguish broad aggregate benefit from concentrated minority loss.

7.8 Citizen Mini-Publics

Citizen panels simulate sampling noise, information quality, manipulation risk, informed support, benefit estimates, and legitimacy. Panel output can affect default-pass eligibility, active-vote routing, final threshold, agenda priority, or affirmative majority burden of proof.

7.9 Leadership, Direct Democracy, and Judicial Review

Leadership agenda control schedules bills using observable public support, salience, uncertainty, majority-party fit, capture risk, and concentrated-harm risk rather than generated true welfare. Citizen-initiative processes let sufficiently salient, high-support proposals bypass legislative floor blockage through a public referendum. Conference-committee processes revise near-miss bicameral bills toward a bargaining target before chamber revotes. Judicial-review processes can invalidate enacted laws with high rights-like harm, uncertainty, policy shock, or capture risk.

7.10 Alternatives, Package Bargaining, Norm Erosion, and Law Registry

Competing-alternative processes generate same-domain alternatives and select by public benefit, public support, chamber-median proximity, or pairwise majority before final ratification. Package-bargaining processes approximate compromise through side-payment costs, implementation-delay costs, modest policy movement, and harm-reducing package trades. Norm-erosion processes raise procedural delay and reduce public support signals under high contention. Law-registry processes allow enacted laws to persist, be reviewed later, and be renewed, repealed, or partially rolled back.

8 Experimental Design

The main submitted evidence comes from the v21-paper campaign. That campaign joins broad assumption cases, adversarial proposal-generator cases, party-system sensitivity cases, and a stylized rising-contention timeline in one artifact. It is now breadth-first: it includes the stylized U.S.-like benchmark, conventional affirmative baselines, leadership agenda control, committee-first regular order, cloture with conference and judicial review, parliamentary-style coalition confidence, direct citizen initiatives, non-default policy tournaments, citizen assembly review, public-interest screening, agenda lotteries, quadratic attention budgets, proposal bonds, harm-weighted thresholds, compensation amendments, package bargaining, public objection, law-registry review, anti-capture safeguards, adaptive risk routing, norm erosion, and a small default-enactment stress-test family. Adversarial generator cases test high-benefit extreme reform, popular harmful bills, moderate capture, delayed-benefit reform, concentrated rights-like harm, and anti-lobbying backlash. Earlier v17–v20 campaigns remain historical development artifacts, including deeper default-pass sweeps, but are not mixed into the main paper evidence base.

9 Assumptions and Limitations

The policy space is one-dimensional. Public benefit is generated rather than empirically estimated. Legislators are synthetic and do not represent named real officials. Lobby groups are abstract collective actors. The calibration layer

screens plausible ranges but does not yet fit raw datasets automatically. Leadership, initiative, conference, judicial, and norm-erosion processes are stylized mechanism prototypes rather than detailed institutional models. Institutional mechanisms are simplified so they can be compared in bundles without modeling the full legal, administrative, judicial, media, and electoral environment. Package bargaining is a one-dimensional proxy for side payments, delays, exemptions, and harm reduction, not a full model of omnibus bargaining or multidimensional coalition trades.

10 Reproducibility

Run tests with `make test`. Run the current campaign with `make campaign` or `make campaign-v21-paper`. Run empirical benchmark screening with `make calibrate`. Run multi-seed checks for all paper scenarios with `make seed-robustness-check`. Run raw-data readiness checks with `make validation-readiness`, and compute raw-data summaries from any local optional inputs with `make empirical-validation`. Run the breadth-catalog family champion screen with `make family-champions` and document catalog composition with `make catalog-breadth`. Build the paper and appendix PDFs with `make paper`. Build a double-blind supplement with `make supplement-anonymous`. After review, generate a non-anonymous provenance manifest with `make public-provenance`. The review package includes generated campaign reports and PDFs; public repository identifying details are withheld during double-blind review and can be restored after review if permitted.

Use of Large Language Model Tools

The research question and core simulator idea originated with the human author. Large language model tools were used for software implementation assistance and for drafting, editing, and reorganizing manuscript text. The author reviewed and is responsible for all code, experiments, reported results, text, and citations. No large language model is listed as an author.

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